Incident Investigation Report

Sysmon / Wazuh Telemetry Analysis

Host: DESKTOP-5OG7C28 (Windows_VM, 192.168.232.130)
Timeframe: 2026-10-23 ~15:33–15:42 UTC+2
Source: Wazuh alerts and Sysmon EventChannel logs

1. Executive Summary

The telemetry shows two distinct clusters of activity. Benign Windows Defender signature update activity (approximately 15:33) in which MpCmdRun.exe (running as SYSTEM) created temporary executables under C:\Windows\SystemTemp\...MpCommU\. These generated low-severity “Executable dropped in Windows root folder” alerts. User-initiated PowerShell activity by account DESKTOP-5OG7C28\Darragh Casey (approximately 15:37) spawned multiple hostname.exe and whoami.exe processes, performed a file-existence check on C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat, launched that batch file via Start-Process (which created a cmd.exe /c process), and caused PowerShell to write short-lived .ps1 policy-test scripts into the user’s %TEMP% directory. The PowerShell/cmd activity produced the higher-severity alerts. All observed behaviour is consistent with execution of a prepared script located under an AtomicRedTeam directory tree on an isolated laboratory virtual machine. No network indicators, persistence, or post-execution malicious payloads are present in the supplied data.

Overall confidence: High (for the sequence and actors observed). The isolated-lab context and explicit AtomicRedTeam path strongly support the interpretation that this is controlled testing rather than real-world compromise.

2. Scope of Activity

• Single Windows endpoint (DESKTOP-5OG7C28 / Windows_VM).
• User context: interactive logon of Darragh Casey (LogonId 0xFCD57).
• Process activity limited to Windows Defender update processes, Security Health Host, Windows Update (UsoClient / MoUsoCoreWorker), Edge updater, and the user-driven PowerShell → cmd chain.
• File creations limited to Defender temporary signature files and PowerShell’s standard __PSScriptPolicyTest_*.ps1 files in the user Temp directory.
• No evidence of additional hosts, lateral movement, data exfiltration, or network connections in the supplied logs.

3. Detailed Investigation

Observed facts

1. At 15:33:26–15:33:40, MsMpEng.exe (Windows Defender engine) spawned MpCmdRun.exe with command lines containing SignatureUpdate / SignaturesUpdateService. These processes created files:
• C:\Windows\SystemTemp\2FF1E121-53D1-4C01-B46F-4C7929E1FDE6MpCommU\mpam-bce1d755.exe
• C:\Windows\SystemTemp\7A3631AE-7D14-4AEF-A570-C9BA3BB82A68MpCommU\UpdatePlatform.exe
(Sysmon Event ID 11, user = NT AUTHORITY\SYSTEM).

2. At 15:37:26, multiple powershell.exe processes (user = Darragh Casey) spawned HOSTNAME.EXE and whoami.exe.

3. One PowerShell command line performed:
if (Test-Path "C:\AtomicRedTeam\atomics\..\ExternalPayloads\Scenario_1_script.bat") {exit 0} else {exit 1}

4. Immediately afterward, PowerShell executed:
Start-Process "C:\AtomicRedTeam\atomics\..\ExternalPayloads\Scenario_1_script.bat"
which caused creation of cmd.exe with:
cmd.exe /c ""C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat" "
(Sysmon Event ID 1, parent = powershell.exe).

5. During the same window, PowerShell created two temporary files:
• C:\Users\Darragh Casey\AppData\Local\Temp__PSScriptPolicyTest_qhypv0d4.v5c.ps1
• C:\Users\Darragh Casey\AppData\Local\Temp__PSScriptPolicyTest_5aj3nptp.trx.ps1
(Sysmon Event ID 11).

Inferences (supported by the evidence)

• The Defender file drops are normal signature-update behaviour; the “Windows root folder” alert is a false positive triggered by the temporary path pattern.
• The user-driven activity is the deliberate execution of a batch script that had previously been placed under an AtomicRedTeam directory structure. The existence check followed by Start-Process is characteristic of a test harness.
• The __PSScriptPolicyTest_*.ps1 files are a well-known side-effect of PowerShell’s execution-policy / language-mode probing and are not themselves malicious payloads.
• Because the host is a laboratory virtual machine and the script path explicitly references AtomicRedTeam, the activity is assessed as controlled testing rather than unauthorised compromise.

4. Evidence Analysis

Timestamp (local)	Event	Key fields	Relevance
15:33:40.626	Sysmon 11 (File Create)	Image = MpCmdRun.exe, Target = …\mpam-bce1d755.exe / UpdatePlatform.exe, User = SYSTEM	Triggered Wazuh rule 92217 (level 6). Legitimate Defender update.
15:37:26.065 / .192	Sysmon 11 (File Create)	Image = powershell.exe, Target = …\Temp__PSScriptPolicyTest_*.ps1, User = Darragh Casey	Triggered Wazuh rule 92213 (level 15). Standard PowerShell temp files.
15:37:27.207	Sysmon 1 (Process Create)	Image = cmd.exe, CommandLine = …Scenario_1_script.bat, ParentCommandLine = powershell … Start-Process …Scenario_1_script.bat	Triggered Wazuh rule 92004 (level 4). Direct evidence of script execution.
15:37:26.xxx	Sysmon 1 (multiple)	Image = HOSTNAME.EXE / whoami.exe, Parent = powershell.exe	Enumeration / environment discovery preceding script launch.

All other process creates (SecurityHealthHost, UsoClient, MoUsoCoreWorker, EdgeUpdate, svchost) are unrelated background system activity and do not affect the assessment of the user-driven chain.

5. Timeline of Events

Time (UTC+2)	Event
15:33:25–15:33:26	SecurityHealthHost.exe and MpCmdRun.exe (SignatureUpdate) start under SYSTEM.
15:33:39–15:33:40	MpCmdRun.exe creates two temporary executables under SystemTemp\…MpCommU\ → alerts 92217.
15:37:26.05–15:37:26.11	PowerShell (Darragh Casey) spawns multiple hostname.exe and whoami.exe.

15:37:26.06	PowerShell checks existence of Scenario_1_script.bat and creates first __PSScriptPolicyTest_*.ps1 → alert 92213.
15:37:27.17	PowerShell issues Start-Process on Scenario_1_script.bat; second __PSScriptPolicyTest_*.ps1 created → alert 92213.
15:37:27.20	cmd.exe /c executes Scenario_1_script.bat → alert 92004.
15:39–15:41	Background Windows Update worker processes (UsoClient / MoUsoCoreWorker) – unrelated.

6. MITRE ATT&CK; Mapping

Technique	Tactic	Supporting evidence	Assessment
T1059.001 / T1059.003	Execution	powershell.exe spawning cmd.exe /c with the Scenario_1_script.bat path; parent command line contains Start-Process	Directly observed.
T1105	Command and Control	Wazuh rule 92213 mapping; files created are only PowerShell policy-test scripts, not downloaded payloads	Alert mapping only; not supported by file content or network activity.
T1570	Lateral Movement	Wazuh rule 92217 mapping on Defender temp files	False positive; files are local Defender signature components.
T1033 / T1082	Discovery	Multiple whoami.exe and hostname.exe spawned by PowerShell	Observed.

No evidence supports techniques involving persistence, privilege escalation, credential access, or collection.

7. Overall Investigation Conclusion

The supplied telemetry documents a short-lived, user-initiated execution of a batch script located at C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat from an interactive PowerShell session belonging to Darragh Casey. Preceding reconnaissance (hostname/whoami) and the characteristic PowerShell temporary script files are consistent with a test harness.

Concurrent Windows Defender signature-update activity generated low-severity alerts that are false positives.

Observed facts are limited to process and file-creation events on a single laboratory virtual machine. **Inferences** that the activity is controlled Atomic Red Team testing rest on the explicit path name, the laboratory context, and the absence of any follow-on malicious behaviour.

Confidence: High.

No further malicious activity is visible in the provided data.